

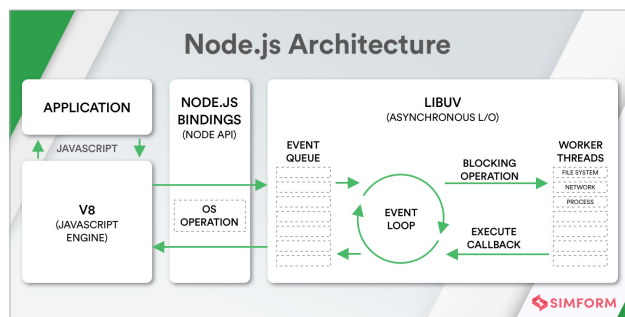


Figure 5: Node.js architecture (Credits: <https://www.simform.com/wp-content/uploads/2020/04/node.js-architecture.png>)

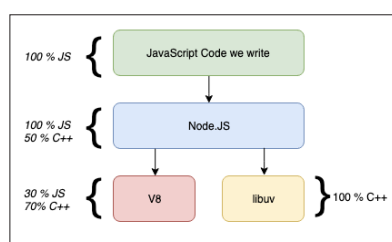


Figure 6: Node.js connected to JavaScript, and V8 and libuv (Credits: <https://easyontheWeb.com/Node.js-v8-and-libuv-under-the-hood-for-beginners/>)

Google. The main purpose of this project is to execute JavaScript code outside the browser. It basically compiles JavaScript code directly to native C++ code before executing it.

libuv is a multi-platform, C++ library that provides support for asynchronous I/O based operations such as file systems, networking and concurrency.

Node.js is best used for IO intensive workloads.

JavaScript frameworks

A JavaScript framework is an application framework written in JavaScript for building user interfaces.

A library offers a function to be called by its parent code, whereas a framework defines the entire application design. A developer does not call a framework; instead, the framework calls and uses the code in some particular way. Example frameworks are AngularJS, Ember.js, Meteor.js, Vue.js, etc.

Table 1 compares the features of two popular frameworks backed by giant names.

Transpilers

Transpilers, or source-to-source compilers, are tools that read source code written in one programming language, and produce the equivalent code in another language.

Anyone who's had to deal with browser compatibility issues, knows it's not as simple as writing JavaScript that runs everywhere. That's because every browser uses a different JavaScript engine: Chrome runs V8, Firefox runs SpiderMonkey, and Internet Explorer runs Chakra. Each has different performance characteristics, each implements a different subset of latest ECMA standards features, and each is approaching full compliance with the spec at different rates.

Technology	AngularJS	ReactJS
Developer	Google	Facebook
Technology type	Full-fledged MVC framework written in JavaScript	JavaScript library (view in MVC; requires Flux to implement architecture)
Concept	Brings JavaScript into HTML. Works with the real DOM client-side rendering	Brings HTML into JavaScript. Works the virtual DOM server-side rendering
Data binding	Two-way data binding	One-way data binding
Dependencies	Manages dependencies automatically	Requires additional tools to manage dependencies
Language	JavaScript + HTML	JavaScript + JSX
Suited for	Best for SPAs that update a single view at a time	Best for SPAs that update many views at a time

Table 1

But now we can write our source in the latest ECMA standards, and let a transpiler translate it to an older version such as vanilla ES5 that works in every browser. If you need to support browsers from the last millennium, you can even compile down to ES3.

JavaScript can run on websites to build dynamic pages, on mobile and desktop apps, and now also as a backend server. Seeing the way it has evolved over the past three decades, it still has a great future. And now AI/ML modules are also being built in JavaScript. An example is TensorFlow.js!

To conclude, each JS framework has its own pros and cons, and we need to analyse which one will suit us. At times the vanilla JavaScript works amazingly well. 

References

Wikipedia, Medium.com, among other references

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The author works in a leading bank as AVP. She is a perennial learner and loves the quote: "The more I learn, the more I realise how much I don't know." - Albert Einstein.